INCH-POUND
MIL-DTL-24643/49F(SH)
w/INT. AMENDMENT 1
11 July 2011

USED IN LIEU OF
MIL-DTL-24643/49F
01 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, 1000 VOLTS, TYPE LSTNW

This specification is approved for interim use by the Naval Sea Systems Command. Other activities in the Department of Defense may use this interim amendment or may continue using MIL-DTL-24643/49F.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation (see table I for thickness). Standard identification code shall be applied by Method 1 or Method 3. If identification Method 1 is used, the insulation color shall be black.
Fourth	-	Three conductors cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm well-rounded assembly.
Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/49	Type and size	AWG conductor size	Insulation thickness min. avg. (inch)	Cable jacket thickness min. avg. (inch)	Overall diameter	
					min. (inch)	max. (inch)
-01UN	LSTNW-3	16 (Class B)	0.030	0.040	0.380	0.411
-02UN	LSTNW-4	14 (Class B)	0.030	0.040	0.415	0.449
-03UN	LSTNW-9	10 (Class B)	0.030	0.050	0.578	0.625
-04UN	LSTNW-14	9 (Class B)	0.045	0.050	0.620	0.670
-05UN	LSTNW-23	7 (Class B)	0.045	0.055	0.703	0.760
-06UN	LSTNW-50	3 (Class C)	0.045	0.060	0.869	0.969
-07UN	LSTNW-75	1 (Class C)	0.055	0.070	1.048	1.134
-08UN	LSTNW-100	0 (Class D)	0.055	0.075	1.171	1.266
-09UN	LSTNW-150	000 (Class D)	0.055	0.075	1.401	1.515
-10UN	LSTNW-200	0000 (Class D)	0.055	0.075	1.555	1.669
-11UN	LSTNW-300	300 MCM (Class D)	0.055	0.090	1.843	1.957
-12UN	LSTNW-400	400 (127) Navy std	0.055	0.090	2.089	2.203
-13UN	LSTNW-30	30 (19)	0.045	0.055	0.799	0.865

TABLE II. Electrical details.

Type and size	Conductor resistance (ohms)	Insulation resistance (megohms)	Voltage withstand min. (volts)	Accelerated service loading current (amperes)
LSTNW-3	4.3	6000	4000	NA
LSTNW-4	2.68	5000	4000	NA
LSTNW-9	1.08	3800	4000	NA
LSTNW-14	0.859	4400	5500	90
LSTNW-23	0.543	3700	5500	115
LSTNW-30	0.358	3000	5500	140
LSTNW-50	0.21	2600	5500	185
LSTNW-75	0.134	2600	7000	250
LSTNW-100	0.106	2400	7000	295
LSTNW-150	0.0674	1800	7000	400
LSTNW-200	0.053	1500	7000	460
LSTNW-300	0.0377	1200	7000	590
LSTNW-400	0.0273	1000	7000	750

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table II
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	See table II
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	See table II
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60

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Permanence of printing (insulation), Method 1 only (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Accelerated service, Sizes 14 and larger (see table II for load current)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

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UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSTNW	3	2500
LSTNW	4, 9, and 14	2000
LSTNW	23 and 30	1500
LSTNW	50, 75, 100, and 150	1000
LSTNW	200, 300, and 400	800

AMENDMENT NOTATIONS: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Preparing Activity:
Navy – SH
(Project 6145-2011-025)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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